

P SATHYA NARAYAN

Full Stack Developer ~ Engineer

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SUMMARY

Frontend Developer with expertise in React, and back-end experience in Flask, Node.js, and Spring Boot. Skilled in developing and optimizing web applications. Successfully filed a patent with the IPO, authored two books on ML, and published a research paper.

SKILLS

Languages: Python, JavaScript, Java, SQL, C++, C.

Technologies: React, Flask, SpringBoot, Azure, AWS.

EXPERIENCE

10/2023 – present **Systems Engineer**

Tata Consultancy Services

• Full Stack Developer - Digital Labs

- Spearheaded a dynamic, data-driven **React** dashboards using **Recharts**, providing **real-time business insights** for **TATA Group's core leadership**. These dashboards streamlined manual reporting processes and enhanced **data-driven decision-making**, serving as a key tool for leadership when presenting to stakeholders.
- Engineered an **Azure Function** in **Python** to efficiently extract, preprocess, and restructure large datasets from Excel files, seamlessly integrating with **Azure Cosmos DB** for scalable storage and retrieval.
- **Collaborated** with a team to **develop** the frontend of an application using **React** and **deployed** it on **AWS**, significantly improving performance by reducing load time from **4366ms to 2204ms**.

Full Stack Developer - Innovation Labs

- **Developed a water quality monitoring system**, achieving a **92% reduction in data processing time** (from 2 minutes to under 10 seconds) by implementing innovative solutions within the Innovation Lab & IoT/DE project.
- Enhanced the emotion detection tool's UI by redesigning some components, resulting in an increase in user accuracy and a reduction in response time.
- **Designed and developed a custom document filtering application** for the talent development team, **processing 1 TB of data in under 30 minutes** and achieving **50% data compression**, thus accelerating the onboarding process for new associates.
- **Led a team of web developers** to improve deployment efficiency and code quality for the ILP Bizz Skill Scheduling project, **reducing critical bugs by 30%** through optimizing the Flask application.

3/2023 – 6/2023 **SDE Intern**

Kerala Police & Institute of Human Resources Development

- Engineered robust APIs using **Django** for a high-stakes policing project, seamlessly integrating machine learning models to significantly enhance predictive capabilities and operational efficiency.
- Designed and implemented advanced data visualization components using **Plotly** and **NumPy**, transforming complex datasets into actionable insights that drove strategic decision-making.

PROJECTS

Full Stack

Human Activity Recognition using Smartphone Sensors and Cloud Support

Developed a system to classify human activities using smartphone sensor data, achieving 90% accuracy. Implemented a full-stack solution using Flutter for mobile app development, Firebase for real-time data processing, and Flask for backend API. Applied KNN algorithm for activity classification and leveraged Azure cloud services for scalable computing.

Web-ML

Malayalam OCR: Extracting Text with Ease

Designed and implemented an OCR API specifically for the Malayalam language. Utilized Flask for API development, integrated Pytesseract for OCR functionality, achieving 90% text extraction accuracy. Created a user-friendly front-end interface using React.

EDUCATION

7/2019 - 7/2023

B.Tech Computer Science Engineering

College of Engineering Chengannur

Graduated with a total cgpa of **9.31** and Specialization with Honours in Machine Learning and Minor in Signal Processing